DS	**Session**	Sorting	**Question Information**	Level 1 Challenge 18

Question description

kkalaiselvan is going to behave as a vehicle driver.

he needs to drive an automobile on a track divided into "N" no. of sub-tracks.

The letter "K" is also assigned to you.

i.e. the maximum number of kilometres a vehicle may go on each sub-track.

You can add any unit of Petrol to the automobile if it can't cover a sub-track. The total kilometres your automobile may go increases by one unit for every unit of gasoline added.

Input:

The first line of input contains an integer T denoting the no of test cases. Then T test cases follow. Each test case contains two space separated integers N and K.

The second line of each test case contains N space separated integers (A[]) denoting the distance of each N sub-tracks.

Output:

For each test case in a new line you have to print out the minimum unit of Petrol your car require to cover all the sub-tracks. If no extra unit of petrol is required, print -1.

Logical Test Cases

Test Case 1

INPUT (STDIN)

```
2
5 6
2 5 4 5 2
5 3
1 6 3 5 2
```

EXPECTED OUTPUT

```
-1
3
```

Test Case 2

INPUT (STDIN)

```
2
5 6
12 15 14 15 12
5 3
11 61 31 51 21
```

EXPECTED OUTPUT

```
9
58
```

Mandatory Test Cases

Test Case 1

KEYWORD

```
void sort(int a[],int n)
```

Test Case 2

KEYWORD

```
for(i=0;i<n-1;i++)
```

Complexity Test Cases

Test Case 1

CYCLOMATIC COMPLEXITY

```
5
```

Test Case 2

TOKEN COUNT

```
275
```

Test Case 3

NLOC

```
40
```